

• LIVE + RECORDED • DEEP DIVE

AI & ML Engineering — 12 Weeks

From fundamentals to building and deploying real models — a rigorous, code-first program for people serious about working in AI.

PRICE	DURATION	LEVEL	FORMAT	SCHEDULE
₹9,999	12 Weeks	Intermediate	Live + Recorded	Sat · 2–5 PM IST



Prasant Mishra

Founder & Instructor, Qurious Academy

What is AI & ML Engineering — 12 Weeks?

This is the program for people who are done being on the outside of AI looking in. Over 12 weeks, you'll go from the mathematical foundations of machine learning all the way to training neural networks, fine-tuning large language models, and deploying production-ready AI systems. Code-first throughout — every concept is implemented, not just explained.

The program is structured in four phases: ML Foundations (statistics, regression, evaluation), Core Algorithms (trees, SVMs, clustering, dimensionality reduction), Deep Learning (neural networks from scratch, PyTorch, CNNs, RNNs, transfer learning), and Applied AI & Capstone (LLM fine-tuning with LoRA, model deployment, and your capstone project). Each phase builds on the last — by Week 12 you'll have the context to understand why every choice along the way was made.

The capstone project is the centrepiece of Month 3. You'll design, build, and deploy a real AI product — something you would genuinely use or could show to an employer. Weekly assignments with model answers, code reviews, and a structured interview prep module complete the picture.

IS THIS RIGHT FOR YOU?

Who Should Enroll

✓ THIS IS FOR YOU IF...

- ✓ You know Python and basic algebra and want to work professionally in ML/AI — as an engineer, researcher, or applied AI practitioner
- ✓ You've done surface-level AI tutorials and want to go deeper — implementing things from scratch and understanding the maths behind them
- ✓ You want to fine-tune LLMs, train your own models, and deploy them to production — not just call APIs
- ✓ You're serious about a 3-hour Saturday commitment over 12 weeks plus assignment work

✗ THIS IS NOT FOR YOU IF...

- ✗ You don't know Python — learn Python first (see Python Mastery 90 Days) then return to this program
- ✗ You only want to build with AI APIs — the "Build with AI" Weekend Cohort is a better fit
- ✗ You're looking for a pure research/academic program — this is engineering-focused; you'll deploy things, not just study them

Prerequisites: Python basics (variables, loops, functions, classes), algebra-level maths (equations, functions, basic probability). No prior ML experience needed. NumPy familiarity is helpful but not required.

12 Weeks, Four Phases

WEEKS 1-3**ML Foundations**

- ▶ **Statistics & probability for ML** — distributions, expectations, Bayes' theorem, MLE; the statistical intuition that underlies every ML algorithm
- ▶ **Linear & logistic regression from scratch** — implementing both algorithms in NumPy; understanding the loss function, gradient, and update rule
- ▶ **Feature engineering & selection** — how to transform raw data into useful features; normalisation, encoding, and feature importance
- ▶ **Model evaluation** — accuracy, precision, recall, F1, ROC-AUC; overfitting vs. underfitting; cross-validation and the bias-variance tradeoff

WEEKS 4-6**Core Algorithms**

- ▶ **Decision trees & Random Forests** — how tree-based models split data, why ensembles work, and when to reach for gradient boosting
- ▶ **SVMs & kernel methods** — support vector machines, the kernel trick, and maximum margin classification explained geometrically
- ▶ **Clustering — K-Means, DBSCAN** — unsupervised learning; grouping unlabelled data and evaluating cluster quality
- ▶ **Dimensionality reduction — PCA, t-SNE** — reducing high-dimensional data for visualisation, compression, and improved model performance

WEEKS 7-9**Deep Learning**

- ▶ **Neural networks from scratch (NumPy)** — implementing forward pass, loss calculation, backpropagation, and gradient descent entirely in NumPy
- ▶ **PyTorch fundamentals** — tensors, autograd, nn.Module, training loops; the framework used by the majority of production ML teams
- ▶ **CNNs for vision, RNNs/LSTMs for sequences** — architectures for structured data types; training on image and sequence datasets
- ▶ **Transfer learning & fine-tuning** — using pre-trained models as starting points; when and how to fine-tune vs. train from scratch

WEEKS 10-12**Applied AI & Capstone**

- ▶ **LLM APIs & fine-tuning (LoRA)** — working with large language model APIs and fine-tuning open-source models on custom datasets using LoRA efficiently
- ▶ **Model deployment** — wrapping models in FastAPI, containerising with Docker, and deploying to cloud infrastructure
- ▶ **Capstone project** — build your own ML product from data collection through training, evaluation, API, and deployment; mentored sprint
- ▶ **Portfolio, GitHub, interview prep** — presenting your technical work effectively; common ML interview questions and how to approach them

LEARNING OUTCOMES

What You'll Walk Away With

01 Train and evaluate ML models end-to-end

— the complete workflow from raw data to a tested, validated model ready for deployment

02 Build neural networks with PyTorch

— implement and train deep learning models with the industry-standard framework

03 Fine-tune LLMs with LoRA

— adapt large pre-trained language models to specific tasks on consumer hardware efficiently

04 Deploy models to production

— FastAPI + Docker + cloud; the full deployment stack for ML applications

05 Ship a real AI product for your portfolio

— a deployed, demonstrable capstone project that proves your ability to employers

WHAT'S INCLUDED

Everything You Get

✓ **Weekly 3-hour live sessions** — every Saturday 2–5 PM IST for 12 weeks

✓ **All recordings** — every session recorded and permanently accessible

✓ **Weekly assignments with model answers** — practical coding problems that reinforce each week's concepts

✓ **Code reviews** — personal feedback on your assignment submissions

✓ **Capstone project mentorship** — dedicated support from ideation through deployment for your Month 3 project

✓ **Learner's Certificate** — digital certificate of completion from Qurious Academy

✓ **Alumni network access** — lifetime access to the Qurious Academy alumni community

Schedule & Pricing

LIVE SESSIONS

Every Saturday · 2:00–5:00 PM IST

DURATION

12 weeks · 36 hrs live

WEEKLY COMMITMENT

~6–8 hours (inc. assignments)

PRICE

₹9,999

❏ Will I need a GPU for this program?

No — we use Google Colab (free GPU) for deep learning exercises. A regular laptop with stable internet is sufficient for the entire program.

❏ How rigorous is the maths?

We introduce maths where it deepens understanding, but always lead with intuition first. You need algebra-level comfort; calculus and linear algebra are taught as needed within the program.

❏ Is this program suitable for career switchers?

Yes — many alumni have used the capstone project and certificate to transition into ML engineering or data science roles. The interview prep module specifically addresses this.

About Prasant Mishra



Prasant Mishra

Founder & Instructor — Qurious Academy

GenAI Solutions Architect

15+ Yrs Teaching

2500+ Students

60+ Projects Delivered

Prasant has designed and deployed production AI systems across 60+ projects — including model training pipelines, LLM fine-tuning workflows, and ML-powered APIs at scale. This program is a distillation of what he's learned building real systems: the patterns that work, the shortcuts that don't, and the foundations you actually need.

His conviction: the best way to learn ML is to implement it from scratch and then deploy it. Every week of this program reflects that — you don't just understand the algorithm, you've written it and shipped something with it.

QuriousAcademy

Focused, live learning for curious minds. No fluff — just the concepts that matter.

AI & ML Engineering — 12 Weeks

Live + Recorded · Sat 2–5 PM IST · 12 weeks

₹9,999

Enroll at quriousacademy.com

or visit quriousacademy.com/courses

Questions? Reach us at hello@quriousacademy.com

© 2025 Qurious Academy. All rights reserved.